

# Economics

## Higher level

### Paper 3 – Mark Scheme

#### SME Predicted Paper – Moderate

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##### General marking guidance:

- Award marks for correct final answers AND for correct working. Accept alternative valid methods.
- Numerical answers should be given exactly or to 2 decimal places unless otherwise specified.
- For explanations, award marks for each distinct correct point up to the maximum.
- Reward fully and correctly labelled diagrams. Unlabelled or inaccurate diagrams receive partial credit only.
- Accept valid alternative economic reasoning where the logic is consistent.
- For read-off-the-graph answers, accept any value within a reasonable band around the intended figure.

**Question 1 [30 marks]****Q1 (a) (i) — Define price taker [2]**

1 mark: A firm that has no influence over the market price of its product.

1 mark: Must accept the price set by the market because of perfect competition / homogeneous product / many sellers / freedom of entry and exit.

**Q1 (ii) — Short-run profit-maximising output [1]**

$Q \approx 107,000$  kg (where  $MC = MR = P = \$4.50$ ). Accept any answer in the range 100,000–115,000 kg read from Figure 1.

**Q1 (iii) — Short-run economic profit/loss at (a)(ii) [2]**

1 mark: Correct method: Profit =  $(P - AC) \times Q$ .

1 mark: At  $Q \approx 107,000$  and  $AC \approx \$3.17$  (read from Figure 1): Profit  $\approx (\$4.50 - \$3.17) \times 107,000 = \$142,310$ . Accept \$120,000–\$160,000 with valid working. Profit is positive (abnormal profit).

**Q1 (iv) — Long-run equilibrium price and output [2]**

1 mark: Long-run equilibrium price = minimum of  $AC \approx \$3.00$ .

1 mark: Long-run output at min  $AC \approx 80,000$  kg (accept 75,000–85,000 kg).

**Q1 (v) — Movement from short-run to long-run equilibrium [4]**

1 mark: Abnormal profit exists in the short run ( $P = \$4.50 > AC \approx \$3.17$ ).

1 mark: No barriers to entry in perfect competition attract new firms into the industry.

1 mark: Industry supply shifts right; market price falls from \$4.50 towards \$3.00.

1 mark: Firm A reduces output to  $\approx 80,000$  kg (where  $MC = AC = P$ ); in long-run equilibrium only normal profit is earned.

**Q1 (vi) — One characteristic of common access resources [1]**

1 mark: Any one valid characteristic, e.g. rivalrous in consumption (one person's use reduces availability for others); non-excludable (hard to prevent others from accessing); subject to over-use (tragedy of the commons); absence of clearly defined property rights.

**Q1 (vii) — Total cost of price-support scheme [2]**

1 mark: Correct method: Cost = Guaranteed price  $\times$  Quantity purchased.

1 mark:  $\$4.00 \times 400,000,000 = \$1,600,000,000 = \$1.6$  billion.

**Q1 (viii) — Externalities diagram and negative externalities [4]**

1 mark: Correctly drawn and labelled externalities diagram — P on y-axis, Q on x-axis, MPC, MSC, MPB = MSB.

1 mark: Private equilibrium  $Q_m$  (MPB = MPC) and social optimum  $Q^*$  (MSB = MSC) identified;  $Q_m > Q^*$ .

1 mark: Vertical distance MSC – MPC (external cost) clearly shown, linked to deforestation / water use / pesticide run-off / carbon emissions.

1 mark: Welfare-loss triangle between MSC and MPC from  $Q^*$  to  $Q_m$  shaded and identified.

**Q1 (ix) — Indirect tax per household per month [2]**

1 mark: Correct method: Indirect tax = Spending  $\times$  (tax rate / (1 + tax rate)) for tax-inclusive price, OR Spending  $\times$  tax rate if interpreting tax as added on pre-tax base.

1 mark: Most straightforward reading: R\$52.00  $\times$  25% = R\$13.00 per household per month. Accept R\$10.40 if candidate uses 25/125 interpretation (tax already included). Award full marks with shown working.

**Q1 (b) — Policy recommendation: sustainable coffee production [10]**

Level descriptors:

0 marks — No appropriate response.

1–3 marks (Level 1) — Little understanding; little use of terminology; no/incorrect links to data; policy identified but not meaningfully explained or evaluated.

4–6 marks (Level 2) — Some understanding; some relevant economic theory; some reference to data; policy explained but evaluation is limited and may lack balance.

7–8 marks (Level 3) — Clear understanding; accurate use of terminology; good use of data; policy well explained and reasonably evaluated; some balance.

9–10 marks (Level 4) — Thorough understanding; strong use of terminology, theory and diagrams; effective integration of data; policy fully explained, evaluated in balance (costs vs benefits, short- vs long-run, stakeholders), with a supported conclusion.

Indicative policies the candidate could recommend:

- Pigouvian tax on coffee production or on deforestation-linked output to internalise the externality (shifts MPC up toward MSC).
- Targeted subsidy for certified sustainable coffee (e.g. Rainforest Alliance) shifting sustainable-sector supply right.
- Tradable deforestation permits / strict land-use regulation to cap forest clearance.
- Reform of the price-support scheme to be contingent on sustainability standards.
- Public investment in productivity-raising inputs so yields rise without further deforestation.

Evaluation points: stakeholder impact (farmers, consumers, environment, government budget); information problems in setting tax rates; risk of 'leakage' to other producer countries; PED of coffee; practical enforcement; distributional effect on smallholders.

## Question 2 [30 marks]

### Q2 (a) (i) — Real GDP for Ethiopia 2021 [2]

1 mark: Correct method:  $\text{Real GDP} = (\text{Nominal GDP} / \text{GDP Deflator}) \times 100$ .

1 mark:  $(111.30 / 105.30) \times 100 \approx \$105.70$  billion.

### Q2 (ii) — Real GDP per capita 2021 [2]

1 mark: Correct method:  $\text{Real GDP} / \text{Population}$ .

1 mark:  $105,700,000,000 / 120,000,000 \approx \$880.83$  per person.

### Q2 (iii) — Two HDI indicators [2]

1 mark each (max 2): Life expectancy at birth; expected / mean years of schooling (education); GNI per capita (PPP \$).

### Q2 (iv) — One disadvantage of the HDI [2]

1 mark: Identify a valid disadvantage (e.g. ignores income distribution within a country; excludes environmental sustainability; excludes political freedoms and human rights; equal weighting of components is arbitrary; slow-moving data availability).

1 mark: Brief explanation of why this limits HDI as a measure of economic development.

### Q2 (v) — Two consequences of economic growth for Ethiopia [4]

2 × 2 marks (identification + explanation). Accept any two of:

- Higher average incomes / poverty reduction.
- Higher tax revenue enabling spending on health and education.
- Inflationary pressure if growth exceeds productive capacity.
- Environmental degradation / unsustainable resource use.
- Widening inequality if gains are concentrated.

### Q2 (vi) — Current account balance 2021 [2]

1 mark: Correct method:  $\text{CA} = (\text{Xg} - \text{Mg}) + (\text{Xs} - \text{Ms}) + \text{Net primary income} + \text{Net secondary income}$ .

1 mark:  $(3,625 - 14,198) + (4,350 - 5,700) + (-652) + 5,860 = -10,573 - 1,350 - 652 + 5,860 = -\$6,715$  million ( $\approx -\$6.72$  billion).

### Q2 (vii) — Two implications of a rising current account deficit [4]

2 × 2 marks (identification + explanation). Accept any two of:

- Downward pressure on the currency (depreciation).
- Must be financed by foreign borrowing or asset sales → rising external debt.
- Signal of weak international competitiveness / deindustrialisation risk.
- Loss of confidence → capital-flight risk.
- If imports are capital goods, may support future growth (positive interpretation).

### Q2 (viii) — Primary-sector dependence as a barrier [2]

1 mark: Identify a valid barrier (price volatility; Prebisch–Singer hypothesis / declining terms of trade; low value-added; vulnerability to weather / demand shocks; limited backward linkages).

1 mark: Link to economic growth and/or development clearly explained.

**Q2 (b) — Policy to revitalise Ethiopian economic growth [10]**

Level descriptors:

0 marks — No appropriate response.

1–3 marks (Level 1) — Little understanding; little use of terminology; no/incorrect links to data; policy identified but not meaningfully explained or evaluated.

4–6 marks (Level 2) — Some understanding; some relevant economic theory; some reference to data; policy explained but evaluation is limited and may lack balance.

7–8 marks (Level 3) — Clear understanding; accurate use of terminology; good use of data; policy well explained and reasonably evaluated; some balance.

9–10 marks (Level 4) — Thorough understanding; strong use of terminology, theory and diagrams; effective integration of data; policy fully explained, evaluated in balance (costs vs benefits, short- vs long-run, stakeholders), with a supported conclusion.

Indicative policies: investment in education and healthcare (HDI focus); infrastructure (transport, electrification, logistics); industrial policy to diversify beyond coffee (e.g. apparel industrial parks); microfinance and agricultural productivity programmes; fiscal and exchange-rate reform; attracting FDI with appropriate safeguards; conflict resolution and institutional reform.

Evaluation: fiscal cost / debt sustainability; institutional capacity and corruption risk; time lags; distributional effects; environmental sustainability; vulnerability to primary-commodity price swings.